



Demonstrating Pythagoras' theorem

Task A: Creating triangles from combinations of squares

1) Create a triangle by joining the vertices of the squares with areas of 1 unit², 3 units², and 5 units².

Using a protractor, measure the size of each angle in this triangle.

smallest angle	25°
middle angle	48°
largest angle	107°

2) Explain why you cannot create an equilateral triangle using a combination of the squares you have been given.

There is only a maximum of two copies of each set of congruent squares (squares with the same area), and you need three congruent squares to create an equilateral triangle.

3) Two triangles can be created by joining the vertices of the following:

- squares with areas of 2 units², 5 units², 6 units² (triangle X)
- squares with areas of 4 units², 10 units², 12 units² (triangle Y)

a) Create these two triangles and measure each of their angles with a protractor.

angles of approximately 35°, 64°, and 81° for both triangles

b) Write down what you notice about the angles between each of these triangles. Can you explain your observations?

The triangles are similar to each other as their angles are the same. Did you notice that the squares that make up triangle Y are twice the area of the squares from triangle X?

4) Create a triangle that is similar to a triangle made from squares with areas 21 units², 27 units², 39 units². Which squares did you use?

smallest square	7 units ²
middle square	9 units ²
largest square	13 units ²

5) Create 3 isosceles triangles from the following squares:

- 4 units², 4 units², 12 units² (triangle A)
- 6 units², 6 units², 12 units² (triangle B)
- 1 units², 8 units², 8 units² (triangle C)

a) Measure the largest angle of each triangle with a protractor.

triangle A	120°
triangle B	90°
triangle C	80°

b) Which triangle has as its largest angle:

an acute angle?	C
a right angle?	B
an obtuse angle?	A

6) Complete the table by joining three squares and writing down whether the largest angle is: acute, obtuse, or right. After choosing three squares, can you predict what type of angle you'll get before you join up the squares?

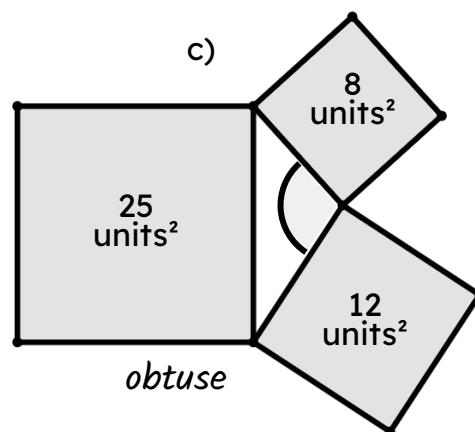
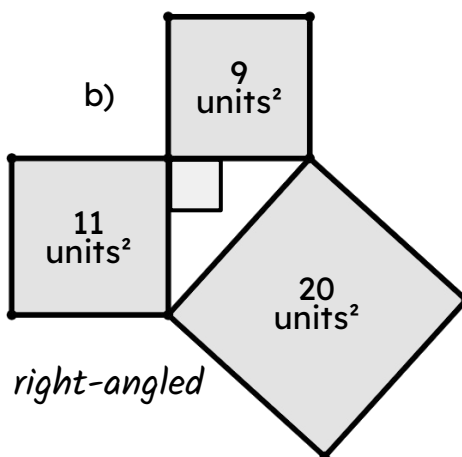
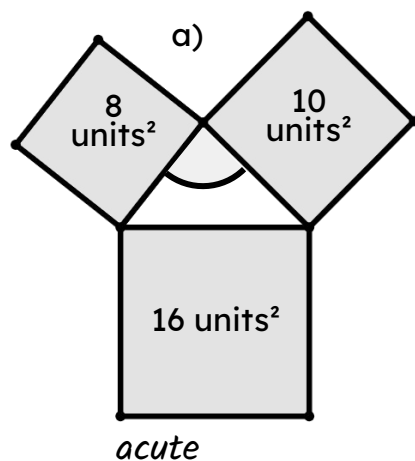
areas of the three squares	largest angle	areas of the three squares	largest angle
7, 10, 13	acute		
4, 5, 11	obtuse		
4, 9, 13	right angle		
3, 10, 13	right angle		
5, 6, 7	acute		



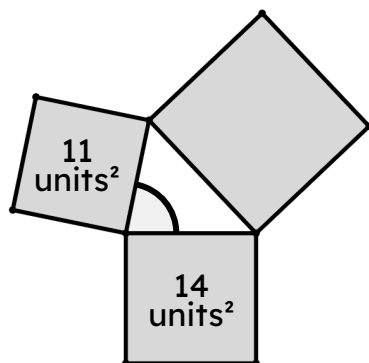
Task B: Properties from summing the areas of squares

1) For each of these triangles, draw on an angle marker at the largest interior angle of the triangle.

For each angle marked, write down whether it is acute, right, or obtuse.

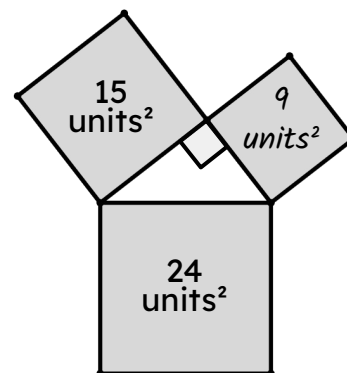


2) The largest angle in this triangle is marked with an angle marker and is acute. Write down a possible area of the largest square.



Any area greater than 14 units², but less than 25 units².

3) The largest angle in this triangle is a right angle. Write down the area of the unlabelled square.



4) Each group shows the areas of three squares whose vertices are joined to create a triangle. Place each group into the correct part of the table that describes the size of its largest angle.

largest angle is acute	largest angle is obtuse	largest angle is right
16, 18, 20	9, 13, 26	8, 13, 21
3, 3, 5	1, 1, 3	8, 17, 25
4, 7, 10	16, 17, 40	9, 16, 25
	14a, 19a, 44a	5x, 7x, 12x

